Transformer Working Principle

1. Introduction

- A transformer is a static electrical device which transfers AC power from one circuit to another at the same frequency, with a change in voltage and current levels, through the phenomenon of mutual induction.
- It has no moving parts and provides electrical isolation between the primary and secondary windings.
- It works on the principle of Faraday's law of electromagnetic induction.

2. Basic Construction

- It consists of two windings (primary and secondary) wound on a laminated iron core.
- The windings are usually made of copper and are electrically isolated from each other but magnetically linked through the core.
- The core provides a low reluctance path for the magnetic flux.

3. AC Supply to Primary

- When an alternating voltage V_1 is applied to the primary winding (N_1 turns), an alternating current I_1 flows through it.
- This current produces an alternating magnetic flux ϕ in the core.
- The flux varies sinusoidally with time.

$$\phi = \phi_m \sin \omega t$$

4. Mutual Induction

- The alternating flux ϕ links both the primary and secondary windings.
- According to Faraday's law, an emf is induced in each winding.

$$e_1 = -N_1 \frac{d\phi}{dt}$$

$$e_2 = -N_2 \frac{d\phi}{dt}$$

5. Transformer Diagram

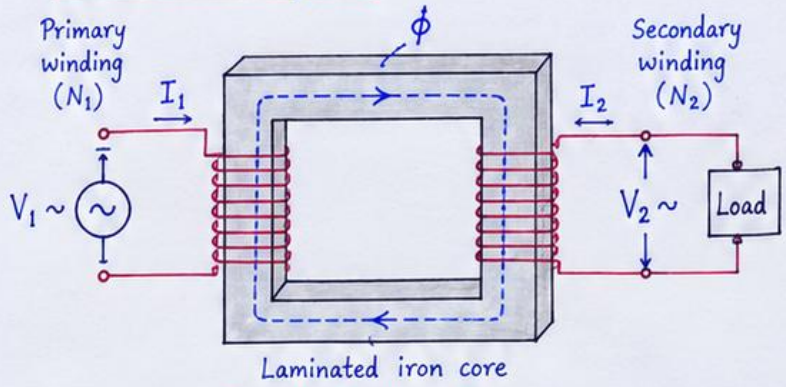
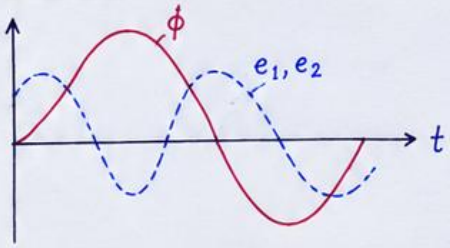


Fig: Basic transformer

6. Waveforms



Induced emf leads the flux by 90°

7. EMF Equations

For a sinusoidal flux, the rms value of induced emf in each winding is

$$E_1 = 4.44 f N_1 \phi_m$$

$$E_2 = 4.44 f N_2 \phi_m$$

where,
- E_1, E_2 = rms emfs (V)
- f = frequency (Hz)
- N_1, N_2 = number of turns
- ϕ_m = maximum flux (Wb)

8. Transformation Ratio

The ratio of secondary voltage to primary voltage is equal to the ratio of turns (for an ideal transformer).

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = k$$

where, k = transformation ratio
- $k > 1$ → Step-up transformer
- $k < 1$ → Step-down transformer
- $k = 1$ → Isolation transformer

9. Key Points

- Transformer works only with AC (not with DC).
- Frequency of secondary voltage is the same as primary frequency.
- Power is transferred from primary to secondary through magnetic flux (no electrical connection between windings).
- In an ideal transformer, input power = output power ($V_1 I_1 = V_2 I_2$).
- In a practical transformer, there are losses (copper loss, core loss, etc.), so output power is slightly less than input power.

EMF Equation Derivation of Transformer

1. Introduction

- A transformer is a static device which transfers AC power from one circuit to another at the same frequency, based on the principle of electromagnetiy induction.
- When an alternating voltage is applied to the primary winding, a time varying magnetic flux is set up in the core, which links both the primary and secondary windings and induces emf's in them.

2. Transformer with N turns

Consider a coil of N turns wound on a magnetic core and linked by a sinusoidal flux ϕ.

3. Flux and its expression

Let the core flux be sinusoidal and given by

$$\boxed{\phi = \phi_m \sin \omega t} \quad \ldots (1)$$

where,

ϕ_m = maximum value of flux (Wb)
ω = angular frequency (rad/s)
f = frequency (Hz)
$\omega = 2\pi f$

4. Instantaneous induced emf

According to Faraday's law of electromagnetic induction, the instantaneous emf induced in a coil of N turns is

$$\boxed{e = -N \frac{d\phi}{dt}} \quad \ldots (2)$$

Substituting (1) in (2),

$$e = -N \frac{d}{dt} (\phi_m \sin \omega t)$$

$$= -N \phi_m \omega \cos \omega t$$

$$= N \phi_m \omega \sin (\omega t - 90°) \quad \ldots (3)$$

Thus, the induced emf is sinusoidal and lags the flux by 90°.

5. Maximum value of induced emf

From (3), the maximum value of induced emf is

$$\boxed{E_{max} = N \phi_m \omega} \quad \ldots (4)$$

11. Transformer diagram

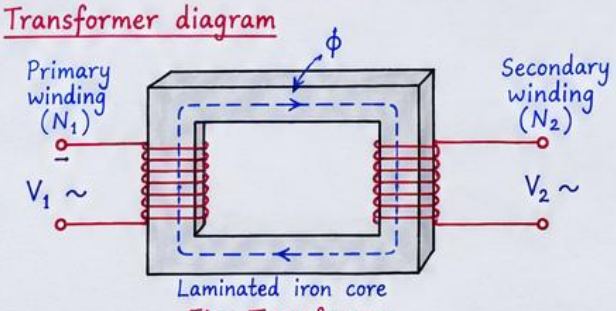
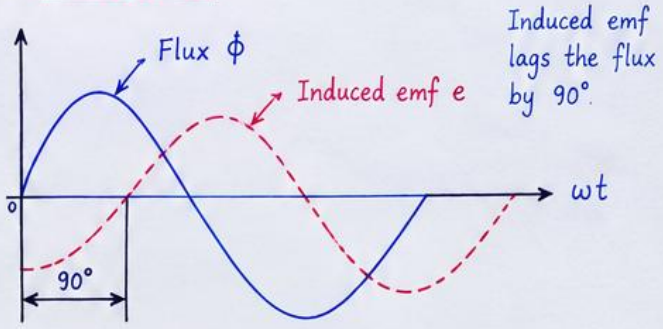


Fig: Transformer

6. Waveforms



7. RMS value of induced emf

The rms value of a sinusoidal quantity is given by

$$E = \frac{E_{max}}{\sqrt{2}}$$

Substituting from (4),

$$E = \frac{N \phi_m \omega}{\sqrt{2}} = \frac{N \phi_m (2\pi f)}{\sqrt{2}}$$

$$\boxed{E = 4.44 f N \phi_m} \quad \ldots (5)$$

8. EMF equations of transformer

For the primary and secondary windings with N_1 and N_2 turns respectively,

$$\boxed{\begin{aligned} E_1 &= 4.44 f N_1 \phi_m \\ E_2 &= 4.44 f N_2 \phi_m \end{aligned}} \quad \ldots (6)$$

where,

E_1 = rms value of induced emf in primary winding (V)
E_2 = rms value of induced emf in secondary winding (V)
f = frequency (Hz)
N_1, N_2 = number of turns in primary and secondary windings
ϕ_m = maximum value of flux (Wb)

9. Important points

1) The emf per turn is $4.44 f \phi_m$.
2) Emf induced in both primary and secondary is proportional to frequency, number of turns and maximum flux.
3) The induced emf lags the flux by 90°.
4) The emf equation is valid for sinusoidal flux.
5) It is independent of core material and winding resistance.

10. Final form

$$\boxed{E = 4.44 f N \phi_m} \quad \text{(rms value)}$$

Transformer Equivalent Circuit and Vector Diagram on No-Load

1. Introduction

- A transformer is a static device which transfers AC power from one circuit to another at the same frequency through the phenomenon of mutual induction.
- When the secondary is open ($I_2 = 0$), the transformer operates on no-load.
- A small no-load current I_o flows in the primary winding to establish the mutual flux in the core and to supply the core losses.
- The no-load current is highly inductive and is only 2% – 5% of full-load current.

2. Equivalent Circuit on No-Load (Exact)

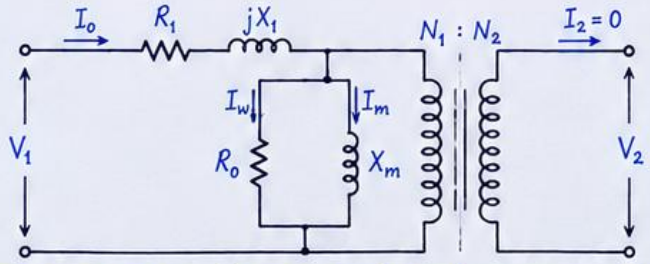


Fig. 1 Exact equivalent circuit of transformer on no-load (referred to primary side)

3. Approximate Equivalent Circuit on No-Load

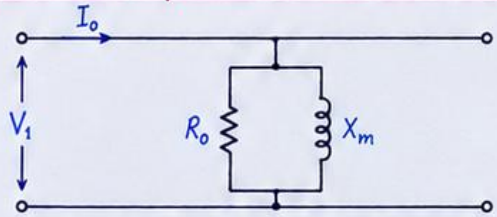


Fig. 2 Approximate equivalent circuit on no-load (R_1 and X_1 are neglected since I_o is small)

4. No-Load Current and its Components

The no-load current I_o has two components:

1) I_w – core loss component (in phase with V_1)
2) I_m – magnetizing component (lags V_1 by 90°)

$$I_o = I_w + j\,I_m$$
$$I_w = I_o \cos \phi_o$$
$$I_m = I_o \sin \phi_o$$

ϕ_o = no-load power factor angle (usually 70° – 80° lagging)

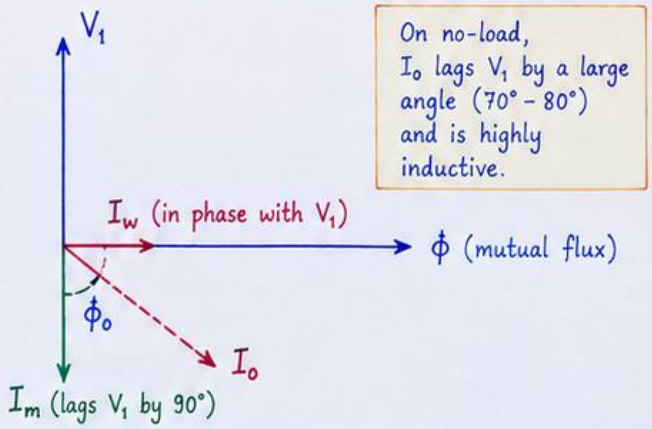
5. Core Loss and Magnetizing Reactance

$$I_w = \frac{V_1}{R_o} \Rightarrow R_o = \frac{V_1}{I_w}$$
$$I_m = \frac{V_1}{X_m} \Rightarrow X_m = \frac{V_1}{I_m}$$

R_o represents core (iron) losses (hysteresis + eddy current losses).

X_m provides the magnetizing current to set up the mutual flux.

6. Phasor (Vector) Diagram on No-Load



On no-load, I_o lags V_1 by a large angle (70° – 80°) and is highly inductive.

Fig. 3 Phasor diagram on no-load

7. Emf and Flux Relations

The induced emf in the windings is given by

$$E_1 = 4.44\, f\, N_1\, \dot{\phi}_m$$
$$E_2 = 4.44\, f\, N_2\, \dot{\phi}_m$$

where,
E_1, E_2 = rms value of induced emf (V)
f = frequency (Hz)
N_1, N_2 = number of turns
$\dot{\phi}_m$ = maximum value of mutual flux (Wb)

8. Relation between V_1 and E_1 on No-Load

$$V_1 = E_1 + I_o (R_1 + j\,X_1)$$

Since I_o is small, the drops I_oR_1 and I_oX_1 are very small.

$$\therefore \quad V_1 \approx E_1 \quad \text{(on no-load)}$$

9. Transformation Ratio

At no-load,

$$\frac{V_2}{V_1} = \frac{E_2}{E_1} = \frac{N_2}{N_1} = k$$

where, k = transformation ratio

10. Key Points

1) No-load current is small (2% – 5% of full-load current).
2) It has two components: I_w (core loss) and I_m (magnetizing).
3) I_w is in phase with V_1 and supplies core losses.
4) I_m lags V_1 by 90° and establishes the mutual flux.
5) The induced emf in both windings lags the flux by 90°.
6) V_1 is approximately equal to E_1 on no-load.
7) R_o and X_m are very large compared to R_1 and X_1.
8) Power drawn on no-load is small and is mainly the core loss.

Conclusion:

On no-load, the transformer draws a small, lagging current to establish the mutual flux and to supply the core losses. The equivalent circuit consists of a shunt branch ($R_o \parallel X_m$) across the supply, and the phasor diagram shows that the induced emf lags the flux by 90° and the no-load current lags the supply voltage by a large angle (70° – 80°).

Ideal Transformer

1. Introduction

- A transformer is a static device which transfers AC power from one circuit to another at the same frequency through the phenomenon of electromagnetic induction.
- An ideal transformer is a theoretical model in which all the losses are assumed to be zero and the windings are perfectly coupled.
- It provides 100% efficient power transfer from primary to secondary.

2. Assumptions of an Ideal Transformer

1) Zero winding resistance ($R_1 = R_2 = 0$)
2) No leakage flux (all the flux links both windings)
3) Infinite core permeability ($\mu \to \infty$, hence magnetizing current is zero)
4) No core losses (no hysteresis and eddy current losses)
5) 100% efficiency ($\eta = 1$)

3. Constructional Diagram

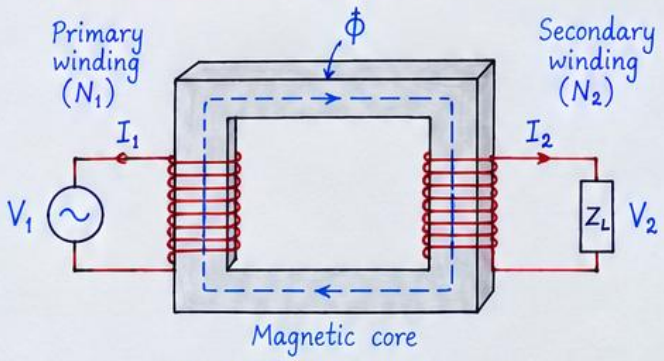


Fig. 1 Ideal Transformer

4. EMF Equations

Since the same flux Φ links both the windings, the induced emfs are

$$E_1 = 4.44\, f\, N_1\, \Phi_m$$
$$E_2 = 4.44\, f\, N_2\, \Phi_m$$

where,

E_1, E_2 = rms value of induced emf in primary and secondary windings (V)
f = frequency (Hz)
N_1, N_2 = number of turns in primary and secondary
Φ_m = maximum value of mutual flux (Wb)

5. Transformation Ratio

From emf equations,

$$\frac{E_1}{E_2} = \frac{4.44\, f\, N_1\, \Phi_m}{4.44\, f\, N_2\, \Phi_m} = \frac{N_1}{N_2} = K$$

$$\boxed{\frac{V_1}{V_2} = \frac{E_1}{E_2} = \frac{N_1}{N_2} = K}$$

where, K = transformation ratio

6. Current Ratio

In an ideal transformer, input power equals output power (no losses).

$$V_1 I_1 = V_2 I_2$$

Using transformation ratio,

$$\frac{I_1}{I_2} = \frac{V_2}{V_1} = \frac{N_2}{N_1} = \frac{1}{K}$$

$$\boxed{\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{K}}$$

7. Phasor Diagram (Ideal Transformer)

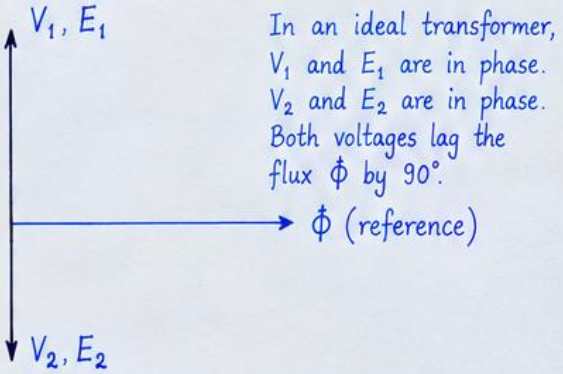


Fig. 2 Phasor diagram on no-load / load (Ideal transformer)

8. Key Points

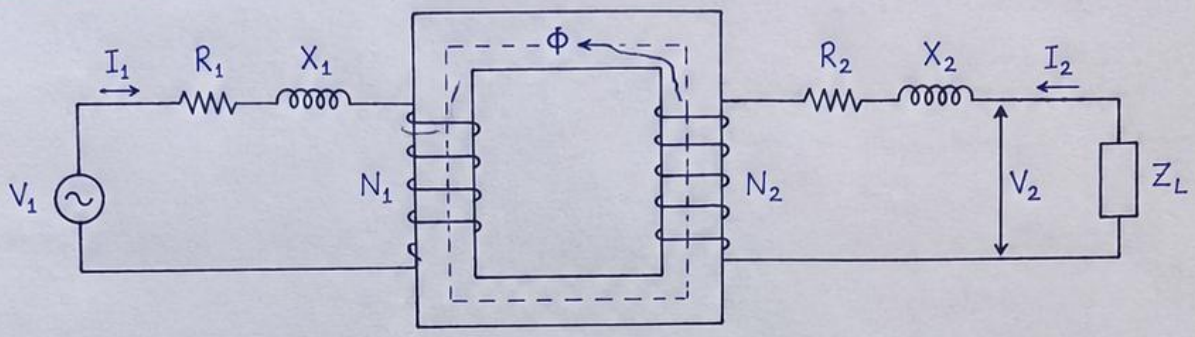
1) All the flux produced by the primary winding links the secondary winding (no leakage flux).
2) No winding resistance, hence no copper loss.
3) Infinite core permeability, hence magnetizing current is zero.
4) No core losses (no hysteresis and eddy current losses).
5) 100% efficiency ($\eta = 1$).
6) $\frac{V_1}{V_2} = \frac{N_1}{N_2}$. and $\frac{I_1}{I_2} = \frac{N_2}{N_1}$.
7) Input power equals output power ($V_1 I_1 = V_2 I_2$).
8) It is an ideal or theoretical model; practical transformers are never ideal.

Real or Practical Transformer Equivalent Circuit

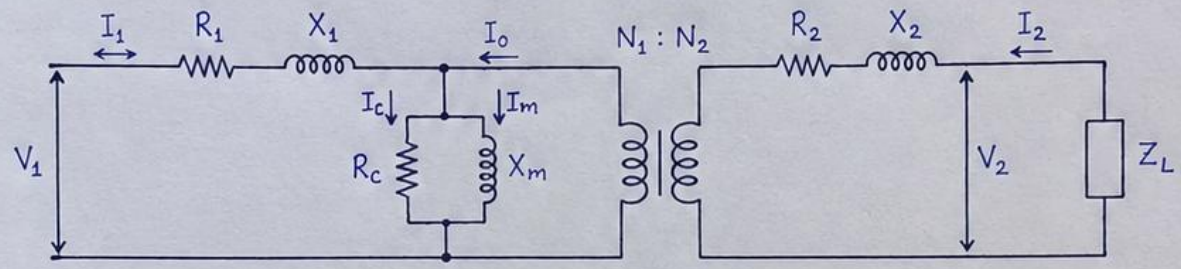
1. Introduction :

In a practical transformer, there are copper losses due to winding resistances and core (iron) losses due to hysteresis and eddy currents. Flux leakage also exists, so the whole flux does not link both the windings. Hence, a practical transformer is represented by an equivalent circuit which includes these non-idealities.

2. Practical Transformer Model :



3. Exact Equivalent Circuit :



4. Elements of the Equivalent Circuit :

1) R_1 - Primary winding resistance (Copper loss in primary)
2) X_1 - Primary leakage reactance (due to leakage flux)
3) R_c - Core loss resistance (represents hysteresis and eddy current losses in the core)
4) X_m - Magnetizing reactance (represents the magnetizing current required to establish mutual flux)
5) R_2 - Secondary winding resistance (Copper loss in secondary)
6) X_2 - Secondary leakage reactance (due to leakage flux)
7) $N_1 : N_2$ - Turns ratio of the transformer
8) Z_L - Load connected to secondary

5. Referred to Primary Side :

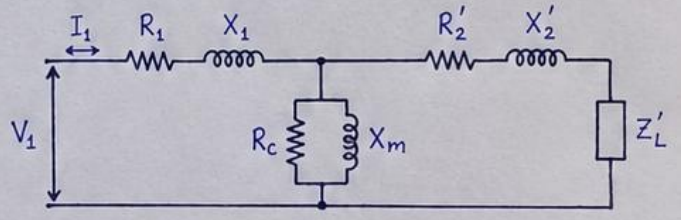
The secondary parameters can be referred to the primary side using the turns ratio

$$a = \frac{N_1}{N_2}$$

$$R_2' = a^2 R_2 \qquad X_2' = a^2 X_2$$

$$Z_L' = a^2 Z_L$$

The equivalent circuit referred to primary side is shown below.



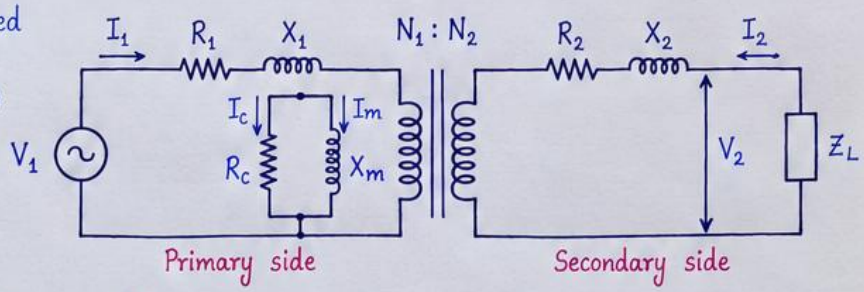
6. Uses :

- To determine voltage regulation
- To calculate efficiency
- To analyse performance under different loading conditions
- To solve problems conveniently by referring all parameters to one side.

Steps to get simplified equivalent circuit of Real or Practical Transformer referred to Primary Side

1. Exact equivalent circuit :

A practical transformer is represented by its exact equivalent circuit with primary and secondary parameters as shown below.



2. Refer secondary quantities to primary side :

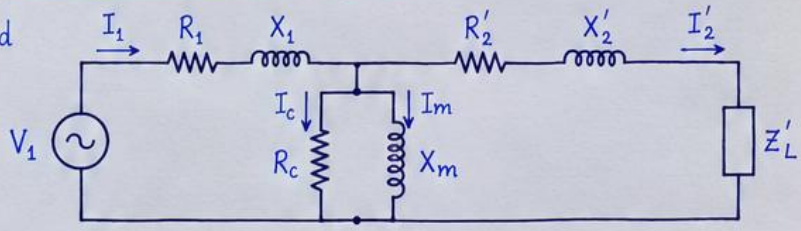
- Let the turns ratio $a = \frac{N_1}{N_2}$.
- Refer the secondary resistance, reactance and load impedance to the primary side.
- The referred values are :

$$R_2' = a^2 R_2$$
$$X_2' = a^2 X_2$$
$$Z_L' = a^2 Z_L$$

where $a = \frac{N_1}{N_2}$

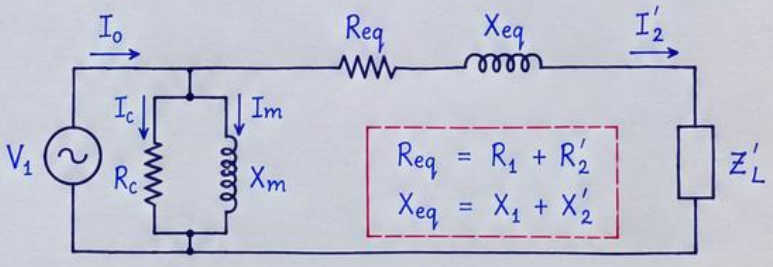
3. Equivalent circuit referred to primary side (without windings) :

- Replace the transformer by the referred secondary parameters. The equivalent circuit referred to primary side is :



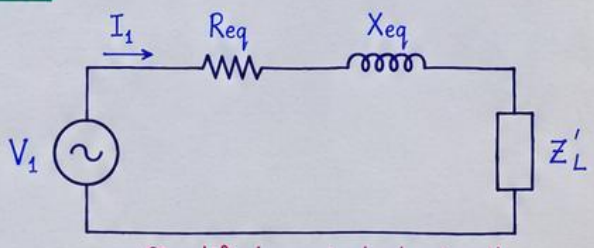
4. Combine the series elements :

- The primary and referred secondary series elements are combined. The parallel branch (R_c and X_m) is kept in parallel with the supply V_1. The equivalent circuit is :



5. Simplified equivalent circuit referred to primary side :

- Since R_c is very large and X_m is very large, the no-load current I_o is small (typically 2-5% of full load current). Hence, for many calculations (voltage regulation, efficiency, etc.), the shunt branch (R_c and X_m) is neglected. This gives the simplified equivalent circuit referred to primary side :



Simplified equivalent circuit (referred to primary side)

6. Final expressions :

$$R_{eq} = R_1 + a^2 R_2$$
$$X_{eq} = X_1 + a^2 X_2$$
$$Z_L' = a^2 Z_L$$

where,

$a = \frac{N_1}{N_2}$

R_1, X_1 – primary winding parameters
R_2, X_2 – secondary winding parameters
R_c – core loss resistance
X_m – magnetizing reactance
Z_L – load impedance

7. Uses :

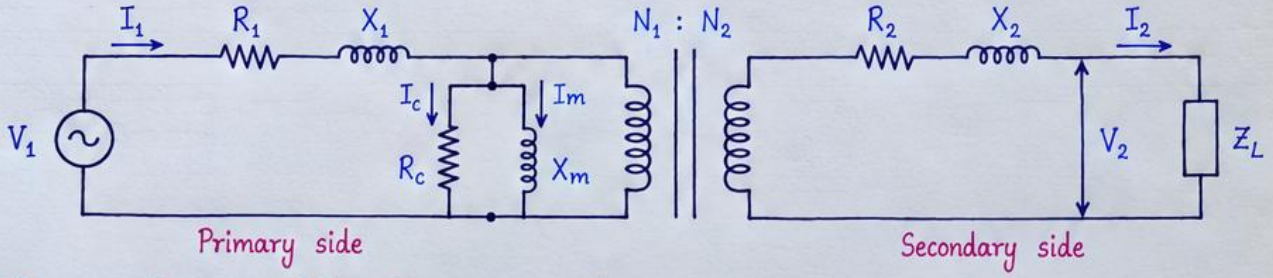
- To determine voltage regulation.
- To calculate efficiency.
- To analyse performance under different loading conditions.
- To make the analysis simpler by referring all secondary parameters to one side (primary side).

Steps to get simplified equivalent circuit of Real or Practical Transformer referred to Primary Side

Date : / /

1. Exact equivalent circuit :

A practical transformer is represented by its exact equivalent circuit with primary and secondary parameters as shown below.



2. Refer secondary quantities to primary side :

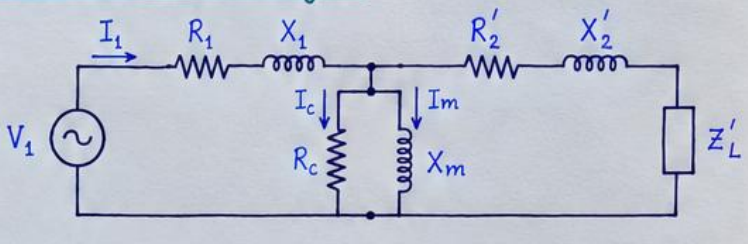
- Let the turns ratio $a = \frac{N_1}{N_2}$.
- Refer the secondary resistance, reactance and load impedance to the primary side.
- The referred values are :

$$R_2' = a^2 R_2$$
$$X_2' = a^2 X_2$$
$$Z_L' = a^2 Z_L$$

where $a = \frac{N_1}{N_2}$

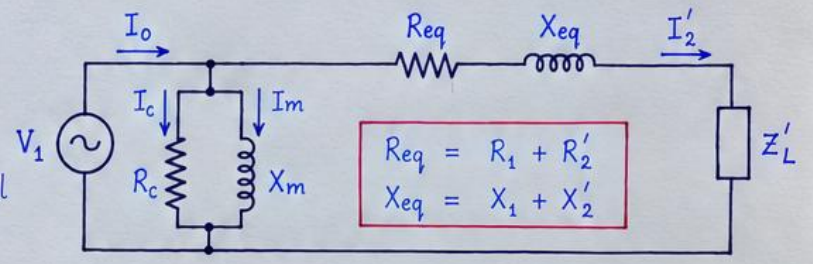
3. Equivalent circuit referred to primary side (without windings) :

- Replace the transformer by the referred secondary parameters. The equivalent circuit referred to primary side is :



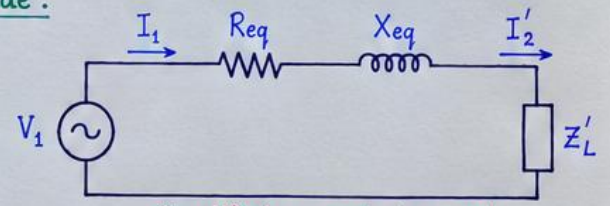
4. Combine the series elements :

- Combine the primary and referred secondary series elements. R_1 and R_2' are in series, and X_1 and X_2' are in series. The shunt branch (R_c and X_m) is kept in parallel with the supply V_1. The equivalent circuit is :



5. Simplified equivalent circuit referred to primary side :

- Since R_c is very large and X_m is very large, the no-load current I_o is small (typically 2-5% of full load current). Hence, the shunt branch (R_c and X_m) is neglected.
- This gives the simplified equivalent circuit referred to primary side :



Simplified equivalent circuit (referred to primary side)

6. Final expressions :

$$R_{eq} = R_1 + a^2 R_2$$
$$X_{eq} = X_1 + a^2 X_2$$
$$Z_L' = a^2 Z_L$$

where,

$a = \frac{N_1}{N_2}$

R_1, X_1	– primary winding parameters
R_2, X_2	– secondary winding parameters
R_c	– core loss resistance
X_m	– magnetizing reactance
Z_L	– load impedance

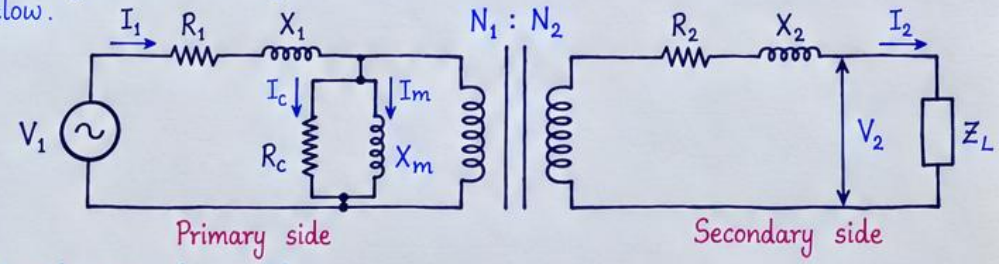
7. Uses :

- To determine voltage regulation.
- To calculate efficiency.
- To analyse performance under different loading conditions.
- To make the analysis simpler by referring all secondary parameters to one side (primary side).

Steps to get simplified equivalent circuit of Real or Practical Transformer referred to Secondary Side

1. Exact equivalent circuit :

A practical transformer is represented by its exact equivalent circuit with primary and secondary parameters as shown below.



2. Refer primary quantities to secondary side :

- Let the turns ratio $a = \frac{N_1}{N_2}$.
- Refer the primary resistance, reactance and shunt branch to the secondary side.
- The referred values are :

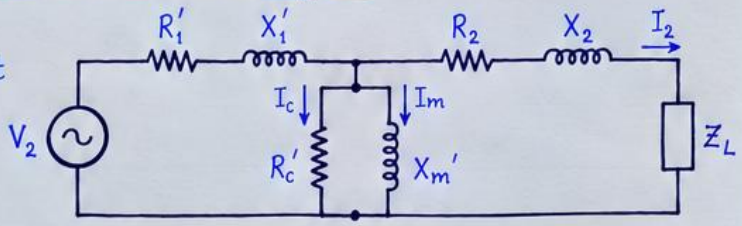
$$R_1' = \frac{R_1}{a^2} \qquad X_1' = \frac{X_1}{a^2}$$

$$R_c' = \frac{R_c}{a^2} \qquad X_m' = \frac{X_m}{a^2}$$

where $a = \frac{N_1}{N_2}$

3. Equivalent circuit referred to secondary side (without windings) :

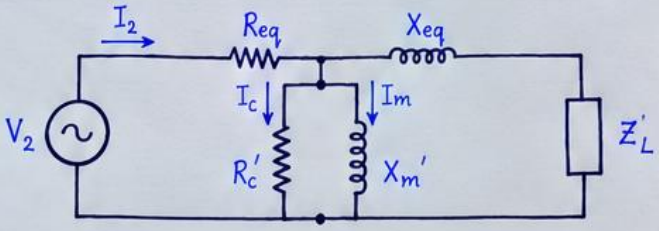
- Replace the transformer by the referred primary parameters. The equivalent circuit referred to secondary side is :



4. Combine the series elements :

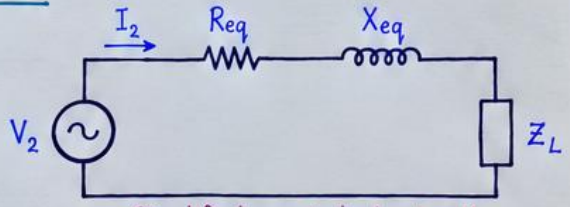
- The referred primary and actual secondary series elements are combined. The equivalent series resistance and reactance are :

$$R_{eq} = R_1' + R_2$$
$$X_{eq} = X_1' + X_2$$



5. Simplified equivalent circuit referred to secondary side :

- Since R_c' is very large and X_m' is very large, the no-load current is small (typically 2–5% of full load current).
- Hence, the shunt branch (R_c' and X_m') is neglected.
- This gives the simplified equivalent circuit referred to secondary side :



Simplified equivalent circuit (referred to secondary side)

6. Final expressions :

$$R_{eq} = R_1' + R_2$$
$$X_{eq} = X_1' + X_2$$
$$R_1' = \frac{R_1}{a^2}$$
$$X_1' = \frac{X_1}{a^2}$$
$$R_c' = \frac{R_c}{a^2} \qquad X_m' = \frac{X_m}{a^2}$$

where,

$$a = \frac{N_1}{N_2}$$

R_1, X_1 – primary winding parameters
R_2, X_2 – secondary winding parameters
R_c – core loss resistance
X_m – magnetizing reactance
Z_L – load impedance

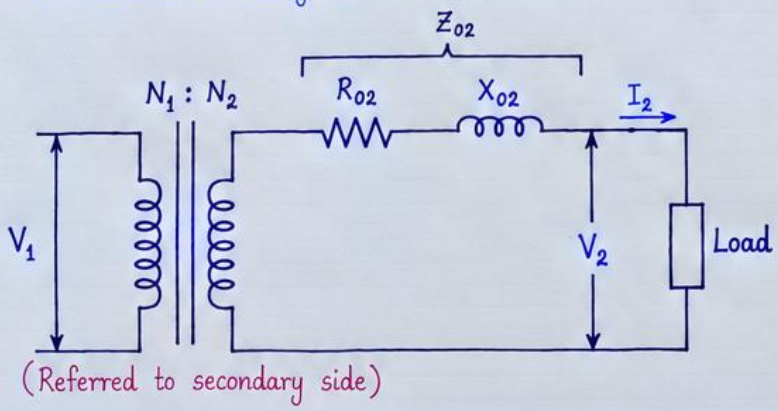
7. Uses :

- To determine voltage regulation.
- To calculate efficiency.
- To analyse performance under different loading conditions.
- To make the analysis simpler by referring all primary parameters to secondary side.

Voltage Regulation of Transformer
(Derivation for lagging p.f.)

1. Simplified equivalent circuit :

The simplified equivalent circuit of a transformer referred to secondary side is shown below.



(Referred to secondary side)

R_{02} – Equivalent resistance referred to secondary side (Ω)

X_{02} – Equivalent leakage reactance referred to secondary side (Ω)

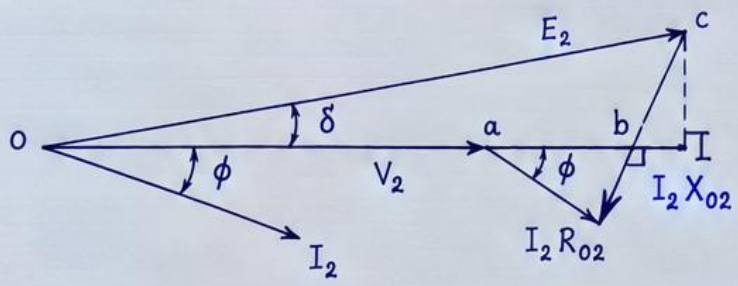
Z_{02} – $R_{02} + j\,X_{02}$

I_2 – Secondary current (A)

V_2 – Secondary terminal voltage at load (V)

E_2 – No-load secondary induced emf (V) (taken as reference)

2. Phasor diagram for lagging p.f. :



In the phasor diagram,
V_2 is taken as reference.
The current I_2 lags V_2 by angle ϕ (for lagging p.f.).
The drop in resistance $I_2 R_{02}$ is in phase with I_2 and the drop in reactance $I_2 X_{02}$ leads I_2 by 90°.
The vector $(E_2 - V_2)$ represents the total voltage drop in the transformer at a given current and power factor $\cos\phi$.

3. Derivation :

From the phasor diagram,

$$E_2 = V_2 + I_2 R_{02} \cos\phi + I_2 X_{02} \sin\phi$$
$$= V_2 + I_2 (R_{02} \cos\phi + X_{02} \sin\phi)$$
$$\therefore \quad E_2 - V_2 = I_2 (R_{02} \cos\phi + X_{02} \sin\phi)$$

4. Voltage Regulation :

Voltage Regulation is defined as the change in secondary terminal voltage from no-load to full-load expressed as a percentage of no-load voltage.

$$\% \text{ Voltage Regulation} = \frac{E_2 - V_2)}{E_2} \times 100$$

Substituting $(E_2 - V_2)$ from above,

$$\% \text{ Voltage Regulation} = \frac{I_2 (R_{02} \cos\phi + X_{02} \sin\phi)}{E_2} \times 100$$

Since at no-load, $V_1 = E_1$ and $V_2 = E_2$, we can write

$$\% \text{ Voltage Regulation} = \frac{I_2 (R_{02} \cos\phi + X_{02} \sin\phi)}{V_2} \times 100$$

5. In terms of primary quantities :

Using $I_2 = k\,I_1$, $E_2 = k\,E_1$ and $R_{02} = k^2 R_{01}$, $X_{02} = k^2 X_{01}$

$$\% \text{ Voltage Regulation} = \frac{I_1 (R_{01} \cos\phi + X_{01} \sin\phi)}{V_1} \times 100$$

where,

$k = \frac{N_2}{N_1}$

R_{01}, X_{01} – equivalent resistance and reactance referred to primary side.